

Compressed-CSI Feedback with Near Real-time Domain Adaptation

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I. SUPPLEMENTARY CLARIFICATIONS AND SUPPORTING RESULTS

A. Environments

We provide the MATLAB code¹ to construct the environments and datasets listed in SUPPLEMENTARY-TABLE I. We use Sionna's default Munich scene to model a 3GPP UMa scenario (E1). We also provide E1 simulation parameters (see SUPPLEMENTARY-TABLE II). For E1, we focus on a single spot: (66 m, 55 m) with UEs uniformly sampled within 20 m radii.

SUPPLEMENTARY-TABLE I
ENVIRONMENT LIST USED IN THIS PAPER FOR CSI DISTRIBUTION AND DOMAIN SHIFTS.

ID	Source	Scenario	Shift driver / observable signature	Type (Reference)
E1	3GPP TR 38.901	UMa	Ray-tracing (RT) in LoS UMa setting [1]. Common, transient perturbations in scattering (e.g., nearby vehicle blockage) may cause CSI distribution shift; often reflected as PDP changes (tap powers / delays). Vehicle placement (main path blockage) realized as perfect electric conductor (PEC) of the size $12 \times 3 \times 4 \text{ m}^3$ near the spot (randomly selected) is considered as domain shifts.	Sionna RT [2]
E2	3GPP TR 38.901	CDL-A to E	Scenario-dependent clustered multipath structure upon non-LoS (CDL-A, CDL-B, and CDL-C) and LoS (CDL-D and CDL-E) environments; distribution shift in PDP can be observed as changes in delay spread (specified values when generating $\mathbf{H}_{\text{org}}$) and cluster/tap power profile (in the range of CDL-A to CDL-E).	Simulation [1]
E3	COST 2100	Indoor / Outdoor	A popular dataset (Indoor / Outdoor) following COST 2100 scenario was provided by [3]. Its shape is $\mathbb{R}^{N_{\text{train}} \times 2N_t N_r N_{\text{tap}}}$ where each value is transformed and clipped to the interval $[0, 1]$ for ease of application to ML-based methods.	Dataset [3]
E4	COST 2100	Indoor / SemiUrban	In this scenario, domain shift is directly characterized by switching the environments; different multipath clustering and PDP shape compared to indoor / semi-urban (outdoor). We consider 2.6 gigahertz environments.	Simulation [4]

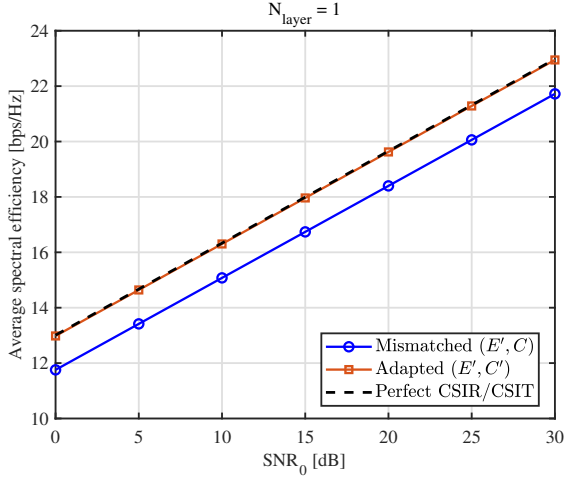
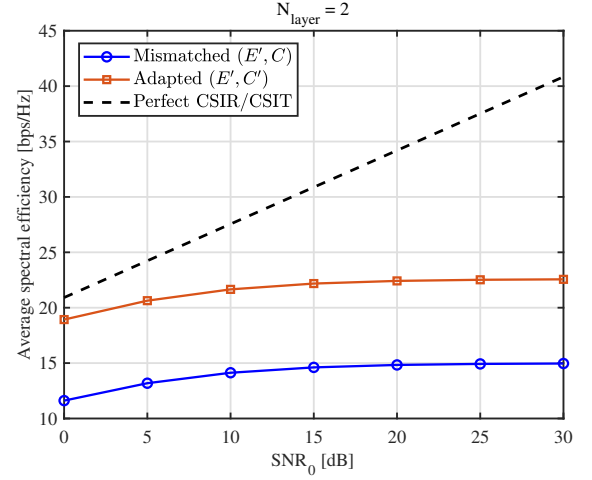
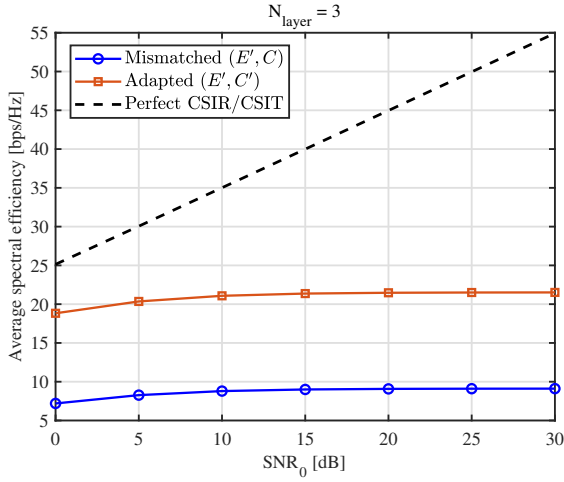
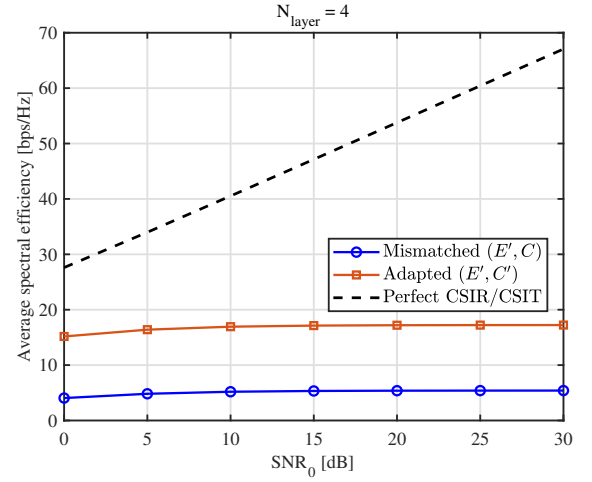
SUPPLEMENTARY-TABLE II
E1 SIMULATION PARAMETERS

Parameter	Symbol	Value	Notes
Carrier frequency	f_c	28 GHz / 3.5 GHz	3GPP TS 38.101-2: FR2 if $> 24 \text{ GHz}$ [5] / added 3GPP TS 38.101-1: FR1 [6]
Bandwidth	B	200 MHz / 5 MHz	FR2 minimum; subcarrier spacing (SCS) $\geq 60 \text{ kHz}$ / FR1 minimum; SCS = 15 kHz
Subcarriers	N_{sub}	250	Number of resource block (in 3GPP) is 264 at most [7]
BS antennas	N_t	10×10	Uniform rectangular array, half-wavelength spacing
UE antennas	N_r	1	Dipole. V-polarization
Channel Taps	N_{tap}	100	Time-domain sparsity
Inter-site distance	ISD	$> 500 \text{ m}$	3GPP UMa default
BS position	P_{BS}	(8.5, 21, 27) m	Rooftop deployment
UE height	h_{UE}	1.5 m	Pedestrian

B. Additional Throughput-oriented Supplementary Results

For completeness, we also provide two additional throughput-oriented supplementary results that support the revised manuscript. First, we report a representative sweep over the data-link SNR SNR_0 to further illustrate its role distinct from $\text{SNR}_{\text{CSI-RS}}$. Second, we provide the full layer-wise throughput table corresponding to the representative cases summarized in the main text.

¹https://github.com/hsjoo-lghtsnd/PSVD_NearRT_DA

(a) E2: CDL-E $\rightarrow$ CDL-A, $N_{\text{layer}} = 1$ (b) E2: CDL-E $\rightarrow$ CDL-A, $N_{\text{layer}} = 2$ (c) E2: CDL-E $\rightarrow$ CDL-A, $N_{\text{layer}} = 3$ (d) E2: CDL-E $\rightarrow$ CDL-A, $N_{\text{layer}} = 4$

S-Fig 1. Representative data-link SNR sweep under post-shift mismatched, adapted, and perfect-CSI settings. The curves illustrate how the realized spectral efficiency varies with SNR_0 while keeping the CSI-acquisition condition fixed, thereby complementing the distinction between $\text{SNR}_{\text{CSI-RS}}$ and SNR_0 used in the revised manuscript.

1) SNR_0 Sweep: S-Fig 1 reports the representative SNR_0 sweep used to complement the revised distinction between $\text{SNR}_{\text{CSI-RS}}$ and SNR_0 .

SUPPLEMENTARY-TABLE III
FULL LAYER-WISE RECONSTRUCTION AND THROUGHPUT RESULTS FOR REPRESENTATIVE E2 AND E4 SHIFTS.

Setting ($E \rightarrow E'$)	N_{layer}	Post-shift mismatched (E', C)		Post-shift adapted (E', C')		R_{perf} [bps/Hz]
		NMSE [dB]	R [bps/Hz]	NMSE [dB]	R [bps/Hz]	
E2: CDL-E $\rightarrow$ CDL-A	1	-0.11	18.40	-14.25	19.62	19.65
	2	-0.11	14.83	-14.25	22.42	34.20
	3	-0.11	9.08	-14.25	21.47	44.97
	4	-0.11	5.38	-14.25	17.20	53.78
E4: Indoor-CSU-2.6G $\rightarrow$ SemiUrban-CSU-2.6G	1	-0.02	15.80	-20.47	21.37	21.38
	2	-0.02	2.91	-20.47	13.61	24.74
	3	-0.02	2.17	-20.47	12.48	25.05
	4	-0.02	1.95	-20.47	11.91	25.05

Notes: The same post-shift CSI reconstruction result is evaluated under different spatial-layer configurations. For each representative shift, the reconstruction NMSE is identical across N_{layer} because the compressed-CSI reconstruction itself is unchanged, whereas the resulting spectral efficiency varies substantially with the layer configuration. These results show that the practical throughput penalty caused by CSI mismatch is highly layer-sensitive, even when the underlying NMSE mismatch is fixed.

2) *Full-layer sweep table*: SUPPLEMENTARY-TABLE III provides the full layer-wise results corresponding to the representative cases summarized in the main text. For both the E2 and E4 shifts, the post-shift mismatched state (E', C') shows a large reconstruction loss relative to the adapted state (E', C') , but the practical throughput degradation is not uniform across spatial-layer configurations. In both cases, the single-layer setting remains relatively more robust, whereas the higher-rank settings are much more sensitive to the same CSIT mismatch. This further supports that system-level evaluation is necessary in addition to reconstruction NMSE when assessing the practical impact of domain shifts in CSI feedback.

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